RELATIVE EQUIVALENCE KERNELS PULL BACK

THE STACKS PROJECT AUTHORS

Editorial note. First displayed isomorphism writes O_X for O_{X_1}, a source typo corrected by context later in the proof; transcribed as-is.

Allowed standard prerequisites (editorial).

- Flat proper base change. perfect-lemma-flat-proper-perfect-direct-image-general, perfect-lemma-compare-base-change (cross-chapter, named). Base change for the cartesian square.
- Tor independence. more-morphisms-lemma-case-of-tor-independence (cross-chapter, named). Verifies the tor independence of the squares.

Source reference. Original definition 0G0I: sources/stacks-equiv/pinned-original-source.tex, source lines 3792–3792. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CTA: focused-seeds/geometry/S9-eqv-base-change-rek/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08IB: focused-seeds/geometry/S9-eqv-base-change-rek/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B91: focused-seeds/geometry/S9-eqv-base-change-rek/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With notation as in Definition 0G0I let K be the Fourier-Mukai kernel of a relative equivalence from X to Y over S. Let $S_1 \to S$ be a morphism of schemes. Let $X_1 = S_1 \times_S X$ and $Y_1 = S_1 \times_S Y$. Then the pullback $K_1 = L(X_1 \times_{S_1} Y_1 \to X \times_S Y)^* K$ is the Fourier-Mukai kernel of a relative equivalence from X_1 to Y_1 over S_1.*

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Proof. Let $K' \in D_{perf}(\mathcal{O}_{Y \times_S X})$ be the object assumed to exist in Definition 0G0I. Denote K'_1 the pullback of K' by $Y_1 \times_{S_1} X_1 \to Y \times_S X$. Then it suffices to prove that we have

$$\Delta_{X_1/S_1,*}\mathcal{O}_X \cong R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* K_1 \otimes^{\mathbf{L}}_{\mathcal{O}_{X_1 \times_{S_1} Y_1 \times_{S_1} X_1}} L\mathrm{pr}_{23}^* K'_1)$$

in $D(\mathcal{O}_{X_1 \times_{S_1} X_1})$ and similarly for the other condition. Since

$$\begin{array}{ccc} X_1 \times_{S_1} Y_1 \times_{S_1} X_1 & \longrightarrow & X \times_S Y \times_S X \\ \downarrow{\scriptstyle \mathrm{pr}_{13}} & & \downarrow{\scriptstyle \mathrm{pr}_{13}} \\ X_1 \times_{S_1} X_1 & \longrightarrow & X \times_S X \end{array}$$

is cartesian it suffices by Derived Categories of Schemes, Lemma 0B91 to prove that

$$\Delta_{X_1/S_1,*}\mathcal{O}_{X_1} \cong L(X_1 \times_{S_1} X_1 \to X \times_S X)^* \Delta_{X/S,*}\mathcal{O}_X$$

This in turn will be true if X and $X_1 \times_{S_1} X_1$ are tor independent over $X \times_S X$, see Derived Categories of Schemes, Lemma 08IB. This tor independence can be seen directly but also follows from the more general More on Morphisms, Lemma 0CTA applied to the square with corners X, X, X, S and its base change by $S_1 \to S$. $\square$